

CLINICAL STUDY PROTOCOL

A Decentralized (Virtual) Randomized Controlled Study

Evaluating a Smartphone-Delivered Cognitive Behavioral Therapy (CBT) Application for Reducing Symptoms of Mild-to-Moderate Generalized Anxiety in Adults

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Week 1 Task: Virtual Study Protocol Development

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1. Introduction and Background

Generalized anxiety disorder (GAD) and subclinical anxiety symptoms are among the most common mental health concerns worldwide, yet access to trained therapists remains limited by cost, geography, and stigma. Digital cognitive behavioral therapy (CBT) applications offer a scalable, low-cost means of delivering evidence-based psychological techniques directly to a participant's smartphone, without requiring in-person clinic visits.

This protocol describes a fully virtual (decentralized) randomized controlled study designed to evaluate whether an 8-week smartphone-delivered CBT program reduces self-reported anxiety symptoms in adults with mild-to-moderate anxiety, relative to a waitlist control group. All study procedures – recruitment, consent, intervention delivery, and outcome assessment – are conducted remotely.

2. Clinical Research Question and Study Objectives

Research Question: Among adults with mild-to-moderate generalized anxiety, does an 8-week smartphone-delivered CBT program, compared with a waitlist control, produce a clinically meaningful reduction in anxiety symptom severity?

2.1 Primary Objective

To evaluate the change in anxiety symptom severity, measured by the Generalized Anxiety Disorder 7-item scale (GAD-7), from baseline to Week 8 in participants using the CBT application compared with the waitlist control group.

2.2 Secondary Objectives

- To assess change in depressive symptoms (PHQ-9) as a secondary outcome.
- To evaluate participant engagement (session completion rate, time spent in-app).
- To assess participant-reported satisfaction and usability of the application (System Usability Scale).
- To evaluate durability of symptom improvement at a 4-week post-intervention follow-up.

3. Study Design Overview

Design: Two-arm, parallel-group, virtual (decentralized) randomized controlled trial.

Setting: Fully remote; no physical study site. All interactions occur via a secure study web portal, the CBT mobile application, video conferencing, and electronic questionnaires.

Duration: 8-week intervention period plus a 4-week follow-up (12 weeks total participation per subject).

Target Sample Size: 120 participants (60 per arm), accounting for an anticipated 20% attrition rate typical of digital health studies.

Allocation: 1:1 randomization to CBT application arm or waitlist control arm, stratified by baseline GAD-7 severity band.

4. Methodology

4.1 Study Population and Recruitment

Participants will be recruited virtually through social media advertisements, university mailing lists, and partnerships with telehealth platforms. Interested individuals will complete an online pre-screening questionnaire hosted on the secure study portal.

4.2 Inclusion Criteria

1. Adults aged 18–65 years.
2. Score of 5–14 on the GAD-7 (mild-to-moderate anxiety range) at screening.
3. Owns a compatible smartphone (iOS 15+ or Android 11+) with reliable internet access.
4. Able to read and provide informed consent in English.
5. Willing to complete all remote assessments for the 12-week study duration.

4.3 Exclusion Criteria

6. Current diagnosis of severe anxiety (GAD-7 ≥ 15), bipolar disorder, psychosis, or active suicidal ideation (screened via item 9 of the PHQ-9).
7. Currently receiving structured psychotherapy (e.g., weekly CBT with a licensed clinician).
8. Initiated or changed psychiatric medication within the past 6 weeks.
9. Current substance use disorder requiring active treatment.
10. Participation in another interventional clinical study within the past 30 days.

4.4 Randomization and Blinding

Eligible, consented participants will be randomized 1:1 using a computer-generated randomization sequence (block randomization, block size 4, stratified by baseline GAD-7 severity band) implemented in the study's electronic data capture (EDC) system. Given the nature of the intervention, participants and the study team cannot be blinded to arm assignment; however, outcome assessments are self-report questionnaires administered through the portal to minimize assessor bias.

4.5 Intervention

Participants in the intervention arm receive access to the CBT application for 8 weeks, comprising weekly self-guided modules covering psychoeducation, cognitive restructuring, behavioral activation, exposure exercises, and relapse prevention. Participants in the waitlist control arm continue usual care and are offered access to the application after their Week 12 follow-up assessment is complete.

4.6 Data Collection Methods

- Screening and baseline data collected via a secure REDCap-based web questionnaire.
- Outcome measures (GAD-7, PHQ-9, SUS) administered electronically at baseline, Week 4, Week 8, and Week 12.
- Application usage/engagement data (logins, module completion, time-in-app) captured passively through in-app analytics.

- One virtual check-in call (video or phone) with study staff at Week 1 and Week 8 to review safety and adherence.
- All data transmitted via encrypted (TLS 1.2+) connections and stored in a HIPAA-compliant, access-controlled cloud database.

4.7 Statistical Analysis Plan

The primary analysis will compare change in GAD-7 score from baseline to Week 8 between arms using a mixed-effects repeated-measures model (MMRM), with treatment arm, time, and their interaction as fixed effects, and baseline severity stratum as a covariate. Analyses will follow the intention-to-treat principle, with missing data handled under a missing-at-random assumption and sensitivity analyses using multiple imputation. Secondary outcomes will be analyzed using analogous mixed models. A two-sided alpha of 0.05 will define statistical significance. Sample size (n=120, 60/arm) provides 80% power to detect a between-group difference of 2.5 points on the GAD-7 (SD=4.5), assuming 20% attrition, at alpha=0.05.

5. Ethical Considerations and Risk Assessment

5.1 Ethics Committee / IRB Approval

This protocol, the informed consent form, and all participant-facing materials will be submitted to an accredited Institutional Review Board (IRB) / Independent Ethics Committee for review and approval prior to any recruitment activity. No study procedures will begin until written approval is obtained, and any protocol amendments will be resubmitted for approval.

5.2 Informed Consent Process

Because the study is fully virtual, consent will be obtained through an e-consent workflow: participants review an electronic consent document, complete a brief comprehension check, and provide an electronic signature. Participants may schedule a live video call with study staff to ask questions before consenting. A copy of the signed consent is automatically emailed to the participant.

5.3 Participant Confidentiality and Data Security

- All participant data will be de-identified using a coded study ID; the key linking ID to identity is stored separately with restricted access.
- Data is encrypted in transit and at rest; the EDC platform is hosted on a compliant, access-logged cloud environment.
- Only authorized study personnel will have access to identifiable data, governed by role-based permissions.

5.4 Risk Assessment and Mitigation

Risk	Likelihood	Mitigation Strategy
Psychological distress from CBT exercises or assessment items	Low–Moderate	Item 9 (suicidal ideation) of PHQ-9 triggers an automated safety alert to study clinician; 24/7 crisis line information provided in-app.

Risk	Likelihood	Mitigation Strategy
Data breach / loss of confidentiality	Low	Encryption, access controls, and routine security audits of the EDC and app platform.
Participant disengagement / attrition	Moderate	Automated reminders, brief check-in calls, and a modest completion incentive.
Technology access barriers (device/internet)	Low–Moderate	Screening confirms device compatibility; technical support desk available throughout the study.
Misclassification due to lack of clinician-administered assessment	Low	Validated self-report instruments (GAD-7, PHQ-9) used; safety items reviewed by clinical staff.

5.5 Data and Safety Monitoring

An independent Data and Safety Monitoring Plan will be maintained by the study coordinator, including automated flagging of any elevated suicidal-ideation item response for same-day clinician follow-up, and a monthly review of adverse event logs by the principal investigator.

6. Study Timeline

Phase	Activity	Duration
Phase 1	IRB submission and approval	Weeks 1–6
Phase 2	Recruitment and screening	Weeks 7–10
Phase 3	Enrollment, consent, and randomization	Weeks 10–12
Phase 4	Intervention delivery (8-week program)	Weeks 12–20
Phase 5	Follow-up assessment (Week 12 post-baseline)	Weeks 20–24
Phase 6	Data cleaning and statistical analysis	Weeks 24–28
Phase 7	Reporting and dissemination	Weeks 28–32

7. Limitations

- Lack of blinding may introduce reporting bias in self-assessed outcomes.
- Fully remote recruitment may limit generalizability to individuals with lower digital literacy or limited technology access.
- Reliance on self-report instruments rather than clinician-administered diagnostic interviews.

8. References

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